

Hierarchical Sparse Conditional Generation of Neutron Energy Deposits in a Zero Degree Calorimeter

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Abstract

We present an auditable development study of a hierarchical conditional generator for raw Geant4 energy deposits from single neutrons in a fixed Zero Degree Calorimeter geometry. The representation has 65 longitudinal layers and 6,790 readout channels, including 2,400 ganged hadronic channels. The generator samples a Bernoulli response branch, total deposited energy, first and active layers, layer budgets, per-layer channel counts, sparse graph support, and positive energy shares. It contains 2,082,507 trainable parameters and was trained on a 551,234-event generator role spanning 0–300 GeV; the reported diagnostic domain is 50–250 GeV. The decoder exactly realizes sampled channel counts and closes to its generated energy budget, yet a repeatedly inspected 10,000-event development bank exposes substantial distributional error. The current checkpoint gives interior inactive layers in 93.6% of nonempty generated events compared with 52.4% in the reference simulation, and 59.0 weak readout-graph components per event compared with 23.2. Its mean number of active channels is 1,520 compared with 1,603. A response-head screen improved its matched scalar loss but generated zero deposit for 50.4% of events, demonstrating that the scalar objective alone was not an adequate promotion criterion. A previously computed shower-only classifier statistic is excluded because its random pair split and omission of the condition do not test conditional fidelity. The versioned artifact reproduces the aggregate figures and documents these stage-level discrepancies; it does not establish physics fidelity, reconstruction equivalence, or simulation speed.

Keywords: calorimeter simulation; neutron shower; sparse generative model; conditional flow matching; graph neural network; reproducibility.

1 Introduction

Geant4 provides detailed particle transport through detector material [1]. Its computational cost motivates learned calorimeter surrogates, but useful acceleration requires accuracy for observables relevant to the intended analysis. Community benchmarks therefore combine conditional response, shower-shape observables, multivariate discrimination, model size, and end-to-end timing [2].

This work studies channelized energy deposits from single neutrons in a fixed Zero Degree Calorimeter (ZDC). The readout is sparse and irregular: one electromagnetic layer has 400 channels, while 64 hadronic layers contain 6,390 channels. Many hadronic readout IDs combine multiple physical locations. We therefore construct a hierarchy that generates the total deposit, longitudinal activity, exact per-layer channel counts, graph-aware support, and energy shares.

Here *sparse* describes exact zeros in the generated readout support. The current implementation still performs substantial dense computation and has no demonstrated computational sparsity advantage.

The contribution is a model specification and a critical development audit. Exact decoder constraints are useful engineering properties, but they do not imply agreement with the reference distribution. We report the current failures directly, identify which measurements are exploratory,

and separate available evidence from experiments that still require event-level data or new simulation. The artifact is a study of raw stored energy deposits rather than a full detector simulator: digitization, photodetector response, electronics, calibration, noise, and reconstruction are outside its target.

2 Related work

Parameterized detector simulation predates current generative methods; AtlFast3 combines parameterized and learned components in an ATLAS production setting [3]. Normalizing-flow approaches include CaloFlow [4], iCaloFlow [5], and layer-to-layer flows [6]. Score and diffusion approaches include CaloScore [7], CaloDiffusion [8], and CaloGraph for irregular geometries [9]. CaloClouds and CaloPointFlow II represent showers as point clouds [10, 11]. CaloDREAM combines conditional flow matching with an autoregressive layer-energy model and transformer-based shower generation [12].

Flow matching has also been applied directly to ALICE ZDC simulation with reported fidelity and inference time [13]. The detector, target, and evaluation protocol differ from those used here, so those measurements provide context rather than a matched baseline. The present architecture combines an irregular readout graph, explicit layer channel counts, exact-size stochastic support, and generated-budget closure. A single checkpoint lineage cannot establish an advantage over alternative representations.

3 Data, geometry, and scope

3.1 Condition and target

The event condition is a single-neutron four-vector $p^\mu = (E_{\text{tot}}, p_x, p_y, p_z)$, in GeV. Kinetic energy is

$$K_{\text{inc}} = E_{\text{tot}} - m_n, \quad m_n = 0.93956542052 \text{ GeV}, \quad E_{\text{tot}}^2 - \|\mathbf{p}\|^2 = m_n^2. \quad (1)$$

The condition encoder receives

$$c_{\text{raw}} = [\log(1 + K_{\text{inc}}/100 \text{ GeV}), u_x, u_y, u_z, \log(E_{\text{tot}}/1 \text{ GeV})], \quad \mathbf{u} = \mathbf{p}/\|\mathbf{p}\|. \quad (2)$$

These five quantities are a deterministic embedding of a mass-constrained four-vector, not five independent degrees of freedom. The minimum audited training kinetic energy is $5.39 \times 10^{-5} \text{ GeV}$, so the direction division is defined for the released training role. Exact zero-energy inputs remain outside the verified encoder domain.

For readout channel i , Y_i is the sum of stored, non-sentinel Geant4 deposited energies mapped to that channel; duplicate records are summed. No additional threshold is applied. The target is $Y = (Y_1, \dots, Y_{6790}) \in \mathbb{R}_{\geq 0}^{6790}$, and $T = \sum_i Y_i$. The production record gives a fixed generator vertex. The study has no material scan or varied vertex, so angular conditioning can be interpreted only within that setup; incomplete upstream-material metadata prevents a stronger acceptance claim.

3.2 Partitions and development sets

The canonical event split is 80/10/10. The training split contains 612,482 events and is partitioned before model fitting into a 551,234-event generator role, a 30,624-event critic-real role, and a 30,624-event critic-monitor role. The checkpoint studied here, V3-SUP at epoch 12 of a planned 24, was initialized from an earlier pilot lineage and then optimized on the full 551,234-event generator role. The 26,624-event subset used for the B0 pilot and short M0/S1/S2/S3 screens is not the training population of V3-SUP.

Training spans $0 \leq K_{\text{inc}} \leq 300 \text{ GeV}$. All diagnostic plots in this paper use $50 \leq K_{\text{inc}} \leq 250 \text{ GeV}$, the declared claim domain. Two fixed development sets are relevant:

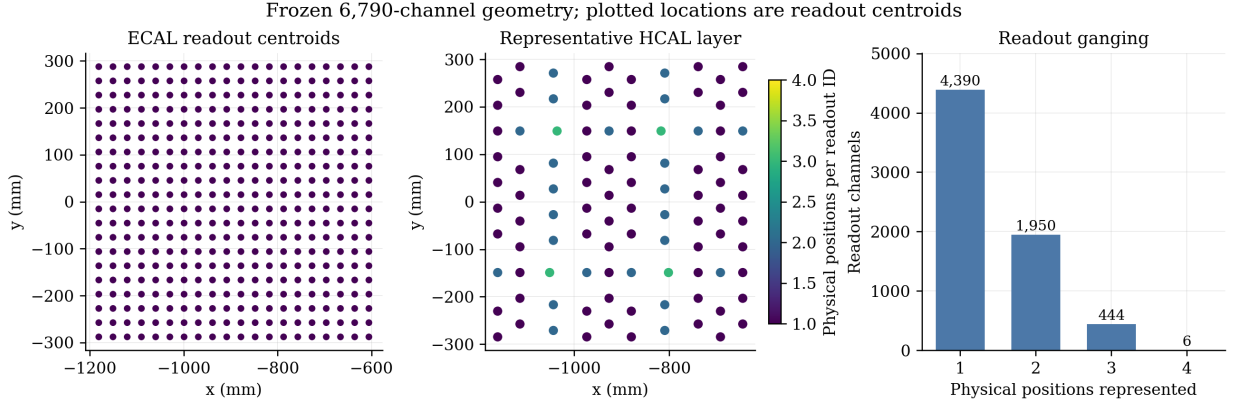


Figure 1: Frozen readout geometry. Coordinates are readout centroids; ganged physical locations are collapsed before modeling. The graph contains directed edges, although the component diagnostic in Sec. 5.1 discards direction.

- D_{bat} : 10,000 conditions and their matched reference events from the canonical validation split, used repeatedly for the diagnostic battery and every plotted comparison;
- D_{loss} : a 4,096-event screen used for component-loss comparisons, with 4,046 positive-deposit events.

The released aggregates do not prove whether these sets overlap, so their statistics are not combined. Repeated inspection makes both development data. No nominal test event contributes to this paper. The nominal test set has also been exposed during earlier project development and cannot serve as a pristine final evaluation set; final claims require a newly locked bank.

3.3 Readout geometry

The frozen geometry contains 6,790 nodes, 65 layers, and 107,920 directed graph edges. Each node connects to its eight nearest readout centroids in the same layer. For each pair of adjacent layers, a node connects to its four nearest centroids in the next layer and the reverse edges are added. Layer 0 has 400 ECAL channels. HCAL layers 1–63 have 100 channels each and layer 64 has 90. A readout coordinate is the unweighted centroid of the stable physical positions sharing its key. Among HCAL channels, 4,390 correspond to one physical position, 1,950 to two, 444 to three, and 6 to four. Thus 2,400 of 6,390 HCAL channels (37.56%) are ganged. The event target and model operate at readout level; separation within a ganged channel is irrecoverable from this representation.

4 Hierarchical generator

4.1 Response branch and zero atoms

The condition encoder maps Eq. (2) to $c \in \mathbb{R}^{128}$. A response head predicts $B \sim \text{Bernoulli}(\sigma(h_B(c)))$ and a four-component Gaussian mixture for

$$z = \log(1 + T/10 \text{ GeV}). \quad (3)$$

At generation time the unconstrained mixture draw z_s is transformed as

$$T_{\text{raw}} = 10 \text{ GeV} \max\{\exp(z_s) - 1, 0\}, \quad C(K_{\text{inc}}) = \min\{0.630110 \max(K_{\text{inc}}, 1 \text{ GeV}), 61.2338 \text{ GeV}\}, \quad (4)$$

$$\tilde{T} = \min\{T_{\text{raw}}, C(K_{\text{inc}})\}, \quad V = B \mathbb{I}[K_{\text{inc}} > 0], \mathbb{I}[\tilde{T} > 0], \quad T = V\tilde{T}. \quad (5)$$

The Gaussian mixture has support at $z_s < 0$; clamping therefore creates a zero atom in addition to $B = 0$. For $K_{\text{inc}} > 0$,

$$P(T = 0 | c) = P(B = 0 | c) + P(B = 1 | c)P(z_s \leq 0 | B = 1, c). \quad (6)$$

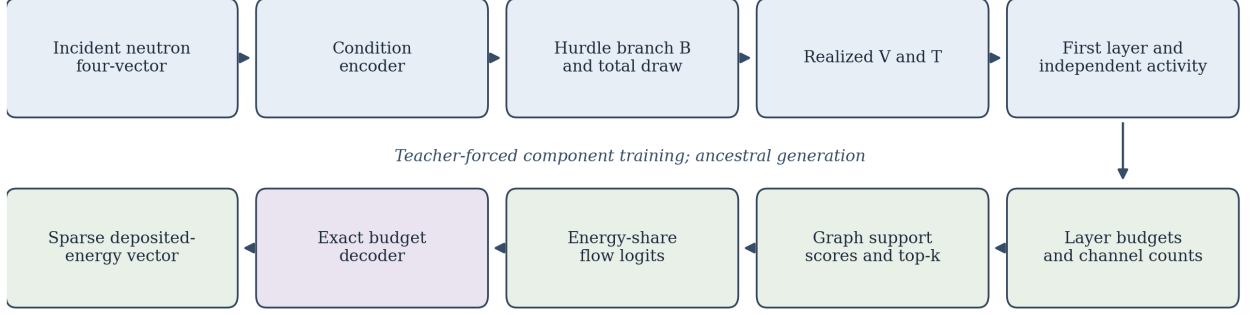


Figure 2: Generation hierarchy. Training uses reference upstream quantities for each downstream loss (teacher forcing). Inference samples ancestrally. The Bernoulli branch B is distinct from realized visibility V , because a Gaussian-mixture draw can map to zero after clamping.

This support mismatch is a limitation of the current head. The observed positive-branch zero count on D_{bat} happened to be zero for all reported checkpoints, but that does not remove the model-level atom. There is no rejection or resampling: the cap is applied deterministically, and all generated events remain in the sample. The available battery did not record how often the cap activated.

For positive reference events the response loss uses a dimensionless measure. Let $E_0 = 1 \text{ GeV}$, $t = T/E_0$, and $s = 10$, so that $z = \log(1 + t/s)$. Since $dz/dt = 1/(s + t)$,

$$-\log p_t(t | c) = -\log p_z(z | c) + \log(s + t). \quad (7)$$

The visible-event binary cross entropy is a separate loss term.

4.2 Longitudinal hierarchy

Given a positive budget, the first active layer is categorical. Subsequent layer activity is modeled as conditionally independent Bernoulli variables given (c, T, L_{first}) . This assumption permits interior inactive layers and does not encode contiguity. It is central to the longitudinal failure observed in Sec. 6.

Positive active-layer fractions are represented in centered log-ratio (CLR) coordinates after clipping shares below 10^{-8} . Conditional flow matching transports independent Gaussian source noise to this target using eight fixed midpoint Euler steps [14]. The source has a component along the additive CLR gauge while the target sums to zero; the terminal softmax removes the gauge from generated fractions. This defines a sampler but not a full-dimensional normalized density on CLR space. The fraction of training shares affected by clipping was not recorded and remains an evidence gap.

4.3 Counts, sparse support, and shares

Each active layer receives a categorical channel-count distribution; inactive-layer zero classes have training weight 0.2. Conditional on the sampled count k_l , a graph network assigns channel logits. Gumbel-top- k_l sampling without replacement selects exactly k_l valid channels [15]. The support network uses three graph blocks, four attention heads, and two bidirectional context layers. The training objective combines class-balanced support BCE, whose positive weight is clamped to $[1, 100]$, and a ranking loss

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \text{softplus}[-(s_i - s_j)], \quad (8)$$

with at most 256 positive-negative pairs per event. This surrogate trains marginal logits and pair ordering; it is not the log likelihood of the exact-size Gumbel sampler. Calibration conditional on k_l has not been measured.

Table 1: Frozen V3-SUP architecture and optimization configuration.

Item	Value
Dimensions	condition 128; shared hidden 96; response 192; profile 128; count 192
Response	four Gaussian components; transform scale 10 GeV
Graph/context	three graph blocks; four heads; two bidirectional context layers; dropout 0
Training	joint, FP32, seed 20260723; 24 planned epochs; epoch-12 checkpoint
Batching	batch 6; gradient accumulation 4; effective batch 24; four workers
Optimizer	AdamW; learning rate 3×10^{-4} , cosine minimum 10^{-6} ; $\beta = (0.9, 0.999)$; $\epsilon = 10^{-8}$; weight decay 0.01; gradient clip 1
Stopping	patience 24; condition encoder trainable; initialized from frozen pilot checkpoint
Flow solver	eight midpoint Euler steps for profile and share flows
Closure	absolute tolerance 2×10^{-5} GeV; relative tolerance 10^{-5}

Positive shares on selected channels use a second clipped-CLR conditional flow with eight midpoint steps. A softmax followed by layerwise rescaling enforces the sampled layer budget. Channels outside support remain exactly zero. The implementation checks finite, nonnegative outputs, valid support, realized counts, and event/layer budget closure. If any invariant fails, the complete generated artifact is excluded from analysis pending diagnosis; events are never silently dropped.

4.4 Architecture and objective

The released V3-SUP configuration retains the v2.2 feature behavior; no optional v3 feature block is enabled. Table 1 gives the exact instantiated configuration. It has 2,082,507 trainable parameters.

The condition network is a 5-to-128 linear projection followed by SiLU, two 128-to-256-to-128 residual MLP blocks, and layer normalization. The response hurdle is 128-to-192-to-1; the mixture path is 128-to-192-to-192-to-12. The first-layer head is 129-to-128-to-65. The activity head combines the condition, log total, and 24-dimensional layer and first-layer embeddings before a 128-unit MLP produces one logit per layer. The count head combines the condition, layer energy, activity flag, and a 24-dimensional layer embedding in a 192-to-192 MLP with 401 output classes, masked to the physical cardinality of each layer. The frozen configuration and pinned source commit define the graph and flow modules exactly.

Training is teacher-forced: each component loss receives reference upstream quantities. Inference is ancestral and can compound upstream errors. An oracle cascade that substitutes reference intermediate states one stage at a time has not yet been run, so the current results do not identify a unique causal component.

The joint loss is

$$\begin{aligned}
\mathcal{L} = & 2.574417\mathcal{L}_{\text{visible}} + 0.160901\mathcal{L}_{\text{response}} + 2.159451\mathcal{L}_{\text{first}} + 0.536770\mathcal{L}_{\text{active}} \\
& + 0.160901\mathcal{L}_{\text{profile}} + 0.160901\mathcal{L}_{\text{count}} + 1.324108\mathcal{L}_{\text{supportBCE}} \\
& + 1.477591\mathcal{L}_{\text{rank}} + 0.444960\mathcal{L}_{\text{share}}.
\end{aligned} \tag{9}$$

Here visible and active are BCE losses; response is Eq. (7); first-layer and count are categorical cross entropies; profile and share are masked conditional-flow-matching MSE losses. The weights were calibrated before the frozen run.

5 Evaluation protocol

For each condition in D_{bat} , the battery compares one stored Geant4 event with one stochastic model draw. This paired design covers 10,000 distinct conditions but does not estimate the distribution at a fixed condition. All intervals and condition-resampling studies therefore remain future work.

5.1 Metrics and conventions

Total-deposit bins use eight 25 GeV-wide intervals from 50 to 250 GeV and include zero-deposit events. The plotted error bars are within-bin sample standard deviations, not uncertainties on the mean. One-dimensional Wasserstein-1 distances are computed exactly from empirical CDFs, but per-bin confidence intervals are unavailable. Response plots are therefore exploratory development diagnostics.

Layer activity and channel support use a strict stored-energy criterion $Y_i > 0$, with no detector threshold. The mean active-layer and active-channel counts include empty events as zero. Gap fraction and mean gap count exclude empty events. A gap is an inactive layer strictly between the first and last active layers. Graph components are computed by unioning the endpoints of the directed edge list; direction is discarded, so the result is a weak-component count. Empty events contribute zero components and zero largest-component fraction. No threshold sweep or independently defined adjacency graph is available, which makes the topology conclusions specific to this representation.

The condition-only control classifier returns AUROC 0.500 because reference and generated samples reuse the same conditions. This is a pipeline sanity check and contains no shower information. A historical classifier was trained only on shower summaries Y and split matched pairs at random across train and test. It is excluded from the manuscript’s quantitative claims because it neither tests the joint distribution (c, Y) nor prevents pair leakage. A valid conditional C2ST must use grouped matched pairs, include raw conditions and shower summaries, disable classifier-internal sample-level validation, and be interpreted against reference–reference, generated–generated, and label-permutation nulls [16]. Those event-level evaluations cannot be reconstructed from the released aggregates.

5.2 Compared checkpoints

The figures show the Geant4 development reference, the earlier B0 baseline, and V3-SUP. M0 is added only where required to interpret the response-head screen S2. B0 and the short screening variants used the 26,624-event pilot subset. V3-SUP continued on the full generator role. Accordingly, B0–V3-SUP differences mix architecture lineage, optimization history, and data exposure; they are descriptive rather than controlled ablations.

6 Development results

6.1 Total deposit and longitudinal profile

V3-SUP has mean total deposit 4.454 GeV on D_{bat} , compared with 4.365 GeV for the reference. Figure 3 shows nonmonotonic bin-level biases from -6.4% to $+6.4\%$ and mismatched widths, with no per-bin uncertainty estimate. The aggregate Wasserstein-1 distance is 0.092 GeV. These values summarize the repeatedly inspected bank and are not a held-out generalization estimate.

The mean longitudinal profile has relative L^1 discrepancy 0.110 for V3-SUP. Its mean ECAL-layer deposit is 2.334 GeV versus 2.100 GeV, an 11.1% excess, while ECAL-start prevalence is lower (91.95% versus 93.81%). These simultaneous shifts indicate that agreement in the total alone can hide compensating errors in where energy starts and how much is placed in the first layer.

6.2 Activity, sparsity, and graph topology

Table 2 and Fig. 5 expose a more substantial structural discrepancy. V3-SUP roughly matches the mean number of active layers but inserts many more interior gaps. It also selects fewer channels and fragments them into more weak graph components.

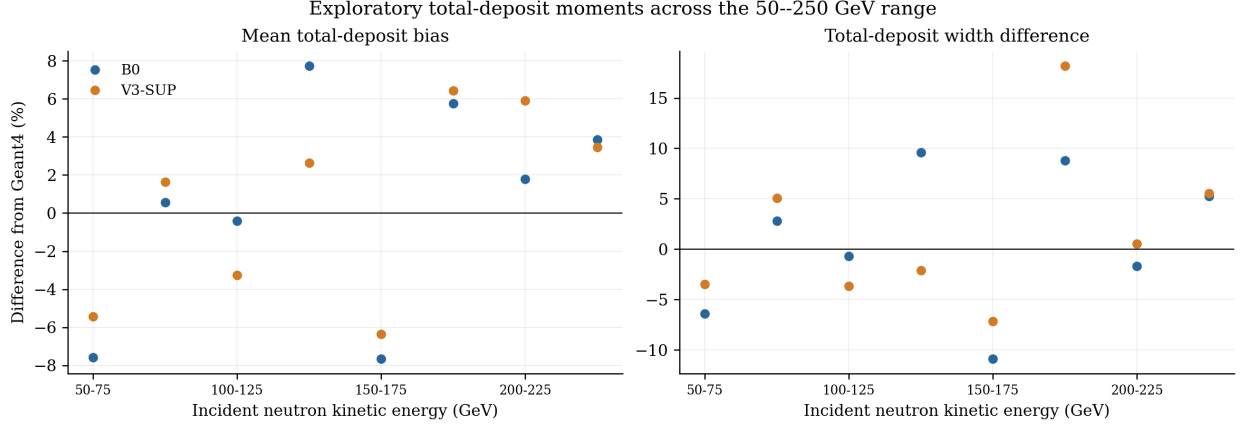


Figure 3: Relative differences in the unconditional within-bin mean and standard deviation of total deposited energy on D_{bat} . Zero-deposit events are included. These are moment comparisons, not confidence intervals.

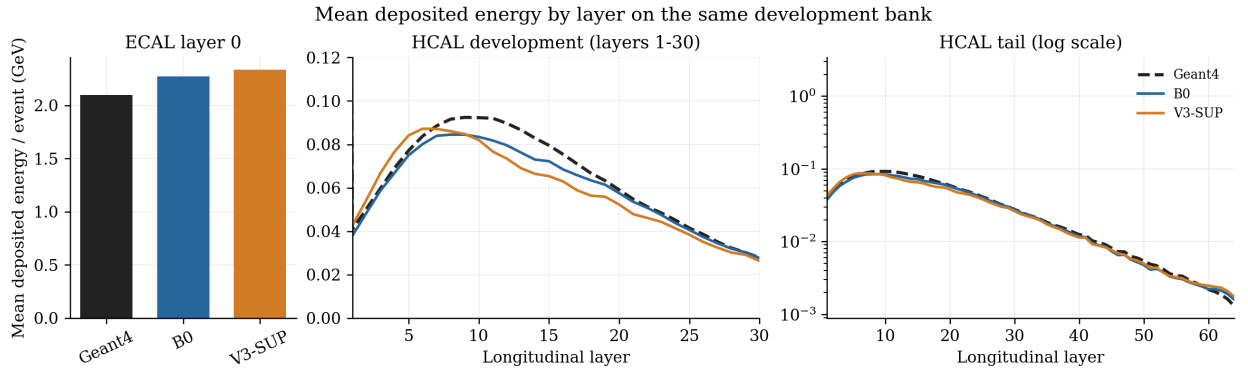


Figure 4: Mean deposited energy by readout layer. The left panel isolates ECAL layer 0, the middle panel shows HCAL layers 1–30, and the right panel shows the HCAL tail on a logarithmic scale.

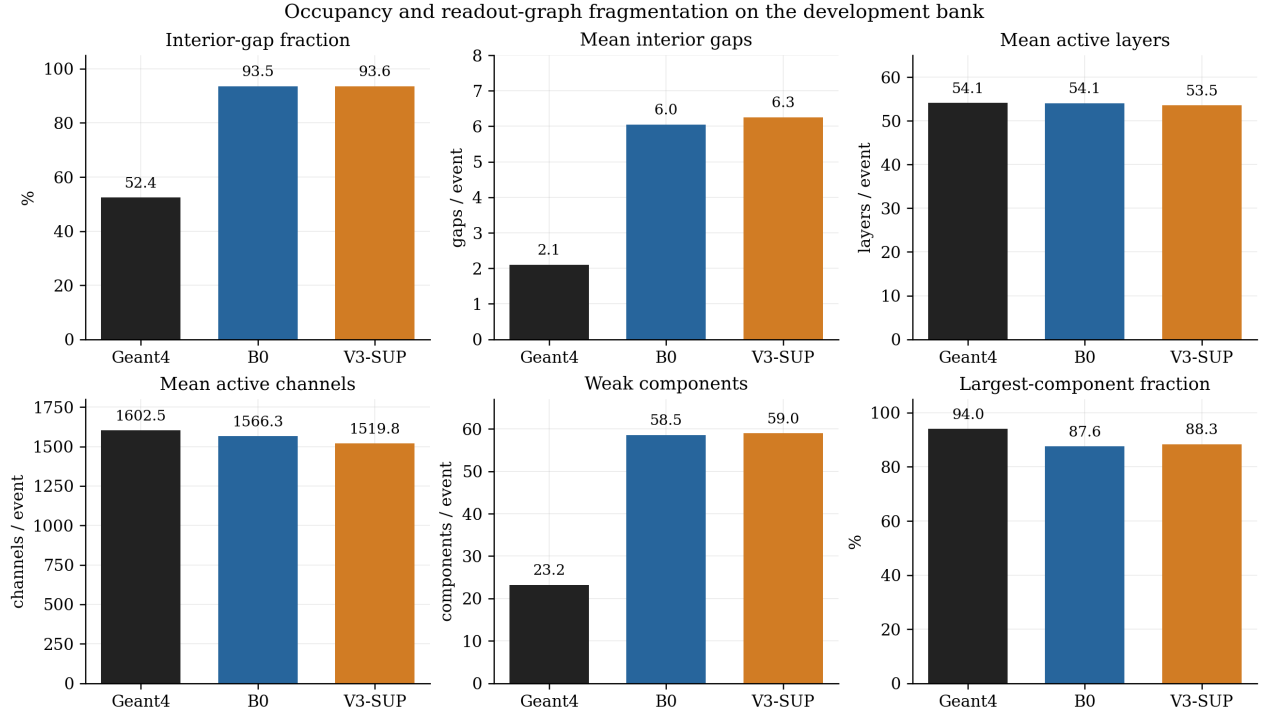


Figure 5: Longitudinal activity, sparsity, and weak graph connectivity. Strict positive stored energy defines activity. The graph panels use the model graph with edge direction discarded.

Table 2: Stage-level diagnostics on D_{bat} . Percentages marked $\dagger$ exclude empty events; other means include them as zero.

Quantity	Geant4 reference	B0	V3-SUP
Zero-deposit events (%)	0.93	1.42	1.61
ECAL-start prevalence (%) $\dagger$	93.81	92.53	91.95
Mean first active layer $\dagger$	0.510	0.735	0.776
Interior-gap events (%) $\dagger$	52.43	93.53	93.56
Mean number of interior gaps $\dagger$	2.10	6.05	6.25
Mean active layers	54.08	54.06	53.53
Mean active channels	1602.5	1566.3	1519.8
Active-channel fraction (%)	23.60	23.07	22.38
Mean weak graph components	23.20	58.55	58.99
Mean largest-component fraction (%)	94.02	87.61	88.33

The direction of these errors is consistent with the independent-Bernoulli activity model and marginal support surrogate, although the absence of an oracle cascade prevents causal attribution.

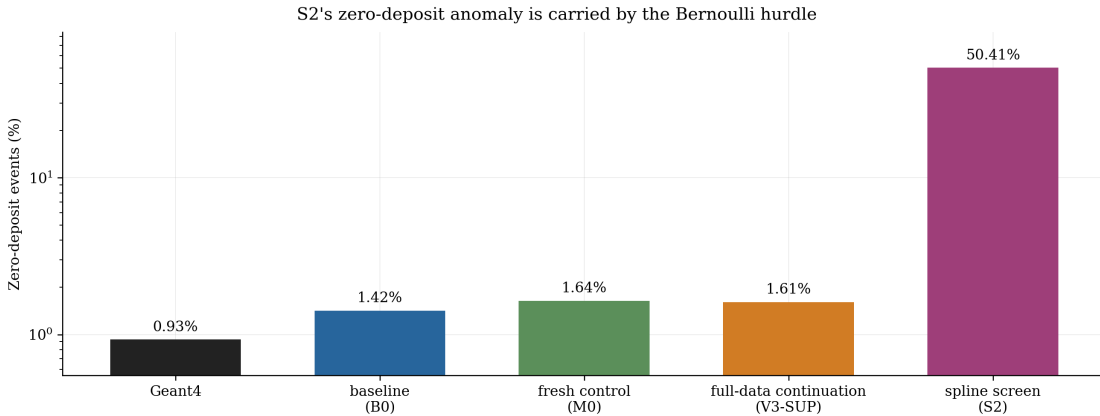
Because 37.56% of HCAL channels are ganged, these component counts describe the readout graph and cannot resolve physical substructure inside a ganged channel. A threshold sweep could also change the number of components sharply. Both qualifications limit any detector-physics interpretation of the absolute values.

The bars report means only. Event-level distributions and occupancy-normalized component measures are not available in the aggregate reports.

6.3 Response-head screen and structural invariants

The S2 response-head screen achieved common-measure loss 4.9212 on D_{loss} , compared with 4.9355 for its M0 control, a decrease of 0.0143. On the separate D_{bat} bank, however, S2 generated zero deposit for 50.41% of events versus 1.64% for M0 and 0.93% for the reference (Fig. 6). All observed S2 zeros came from the Bernoulli hurdle; the positive-branch clamp contributed none on this bank. The available evidence shows that this loss-improving screen had an unacceptable generation anomaly. It does not determine whether the cause was the head, checkpoint state, training dynamics, or another implementation interaction.

The numerical scale of this loss is defined by Eq. (7). On the 4,046 positive events in D_{loss} , the mean Jacobian term is 2.6223 and contributes 0.4219 after multiplication by the response weight. This explains its base measure; it does not make loss values comparable across the two development banks.

**Figure 6:** Zero-deposit fraction on a logarithmic scale. S2 improved the matched scalar loss on D_{loss} but failed the separate generation diagnostic on D_{bat} .

All plotted generated artifacts passed the implemented finite-value, nonnegativity, support-mask, count-realization, and generated-budget checks. For V3-SUP the maximum event closure error was 1.53×10^{-5} GeV, with zero count or support mismatches. The maximum reported layer error, 4.20×10^{-5} GeV, is below the effective tolerance after the relative term is included. These checks establish decoder consistency only; they do not validate the generated distribution.

7 Limitations and required validation

The present evidence records stage-level discrepancies rather than a performance claim or causal diagnosis. The highest-priority missing measurements are:

1. a locked evaluation bank and three independent training seeds for each frozen final condition;
2. a condition-aware, pair-grouped joint C2ST with null calibrations, plus energy and angular stratification;
3. repeated reference and generated showers at fixed conditions to test conditional spread;
4. an oracle cascade for response, first layer, activity, profile, counts, support, and shares;
5. cap-activation and CLR-clipping counters, conditional support calibration, threshold sweeps, and an independent physical-neighbor graph;
6. complete Geant4 version, physics-list, production-cut, geometry, field, vertex, upstream-material, and random-seed provenance;
7. matched implementations of representative dense, point-cloud, graph, and flow-matching baselines; and
8. reconstruction-level observables and end-to-end timing that includes condition preparation, sampling, decoding, transfers, and any postprocessing.

The current battery uses one reference shower and one generated shower per condition, one generator training seed, and no matched timing. It therefore cannot establish tails, fixed-condition uncertainty, angular interpolation, robustness, acceleration, or analysis equivalence.

8 Conclusion

We specified and audited a 2.08-million-parameter hierarchical sparse generator for raw neutron energy deposits in a fixed ZDC readout. The decoder provides exact-size support and generated-budget closure, while the development bank reveals excessive longitudinal gaps, readout-graph fragmentation, reduced channel occupancy, and an ECAL energy excess. A component screen further showed that a slightly improved response loss can coexist with a severe zero-deposit anomaly. We also excluded a historical shower-only classifier result whose split and feature construction did not test conditional fidelity. The versioned aggregate evidence supports a constrained negative finding: strong structural invariants do not guarantee agreement with the conditional Geant4 distribution. The experiments in Sec. 7 are required before this model can support a physics-fidelity or speed claim.

Code and data availability

The manuscript source, frozen portable configuration, aggregate reports, event-independent geometry, deterministic figure builder, and audit records are available at <https://github.com/JulianAttemptsCoding/fast-mc-paper>, release v0.2.0. The model implementation used for this audit is identified by commit f6a3689013d64fb5603f7fe06c9bd9363e4d1d3c of <https://github.com/JulianAttemptsCoding/fast-mc-handoff-2>. The collaboration-owned event file and checkpoints are not redistributed. The public aggregates reproduce every figure and quoted battery statistic, but they are insufficient to retrain the generator or recompute event-level tests.

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Author contributions

Julian Juan conceived the study, developed the software and model, performed the analysis, and wrote the manuscript.

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